



Open-Source, High-Speed Entropic Lattice Boltzmann Flow Solver for Aerospace Problems

Sean Duncan*, Paul Orkwis†

University of Cincinnati, 2600 Clifton Ave, Cincinnati, 45221, Ohio

A new, open-source Lattice Boltzmann Method (LBM) flow solver has been developed at the University of Cincinnati (UC). It applies a powerful, compressible version of the LBM based on entropy with an adaptive mesh refinement. It is capable simulating flows up to Mach 2.5. It shows reasonable agreement with canonical cases. It also shows good agreement with a Mach 1.5 shock generator case and a supersonic exhaust converging-diverging nozzle case.

Nomenclature

P - Pressure
p - Momentum
T - Temperature
 ρ - Density
s - Entropy
f - Distribution function
g - Energy distribution function
c - Lattice velocity, discretized
q - Contracted heat flux
Q - Heat flux or parameter of interest
l - Length
t - Time
r - Refinement ratio
u,v,w - x,y,z velocities, respectively
R - Gas constant
D - Number of dimensions in simulation
 $\mathcal{H}$ - Boltzmann's H
Re - Reynolds number
M - Mach number
 μ - Dynamic viscosity
 ν - Kinematic viscosity
 ε - Energy
 α - Entropic relaxation parameter
 κ - Thermal conductivity
 τ - Mean free time
 δ - Distribution function ratio, Kronecker delta
 γ - Ratio of specific heats
 β - Relaxation parameter
 Ω - Collision operator
 $\zeta, \chi, \lambda, \pi$ - Lagrange multipliers

*PhD Graduate, University of Cincinnati, Department of Aeronautical Engineering and Engineering Mechanics, Member AIAA, currently at Northrop Grumman

†Professor Emeritus, University of Cincinnati, Department of Aeronautical Engineering and Engineering Mechanics, Associate Fellow AIAA

Subscripts & Superscripts

- i, j - $i^{\text{th}}, j^{\text{th}}$ lattice velocity or direction
- α, β, γ - Subscripts denoting direction
- eq - Equilibrium value
- neq or (1) - Non-equilibrium value
- coll - Collision
- latt - Lattice
- wall - Wall direction
- f, c - Fine and coarse, respectively
- ref - Reference value
- * - Nondimensional quantity, quasi-equilibrium value

I. Introduction

A new LBM code for computational fluid dynamics (CFD) has been developed at the University of Cincinnati (UC), the University of Cincinnati Lattice Boltzmann Solver (UCLBS). The code was developed in an effort to have a modern LBM solver capable of solving a large range of problems. The code was created to be open-source to fill a void in the public space since other free or publicly available LBM solvers are limited to low Mach problems.

Three major thrusts have been made to extend the range of LBM flow solvers into transonic and supersonic flow regimes. The first is PowerFlow from the Exa Corporation, now Simulia and part of Dassault Systems, which solves entropy on an expanded lattice, allowing flow up to $M \sim 1.9$ [22]. The second is the work of Karlin et al. at ETH Zurich, who developed a system based upon strict observation of the Laws of Thermodynamics [4–7, 12, 27] called the Entropic Lattice Boltzmann Method (ELBM). The last system relies on a circle function with an extra advection step [9, 37, 38] developed primarily at the Northwestern Polytechnical University and Xiamen University in Xi'an, China.

Traditional Navier-Stokes (NS) flow solvers are often implicit, solving a system of equations. In contrast, the LBM equations are far simpler and algebraic, with no derivatives necessary. However, LBM is almost universally an explicit solution method, though a few implicit methods have been developed [42, 43]. The timestep scale for LBM is generally extremely small as the local timestep scales with local lattice length, $dx \propto dt$. Additionally, large amounts of memory are required as two data points for every direction at every node are required to account for the distribution function, as well as memory for relevant flow variables.

Since its creation, LBM has rapidly grown into a capable simulation technique able to solve many aerospace flow problems including supersonic flows and computational aeroacoustics [17, 18]. Nasa's LAVA-LBM compares favorably with unsteady Reynolds-Averaged Navier-Stokes (URANS) and large-eddy simulation (LES) solvers [25]. LBM is also capable of solving direct numerical simulation (DNS) problems with only minimal addition [33]. Simulia's PowerFlow has been used for many purposes from automotive to aeronautical, including accurate simulation of aeroacoustic problems [14–18]. In general, a steady NS solver will be computationally faster than an LBM solver, but comparable or better speed can be achieved by LBM for unsteady problems.

Adaptive mesh refinement (AMR) is also becoming more popular in LBM solvers. AMR in LBM solvers cannot change grid density or derivative stencil size, so it must increase cell or node density. Update procedures prevent divergence or negative distribution functions. The changing nature of the mesh can balance accuracy, in regions where a finer grid is needed, and computational speed, in regions where the mesh can be coarsened without loss of accuracy. The cost is increased complexity and computational time handling a regridding step. External programs have been developed, such as Amrex [2, 3], which is used in UCLBS, to handle the AMR and leave the creation of the flow solver to the developer.

UCLBS incorporates AMR systems based on user-defined inputs. Additionally, output is performed in parallel using HDF5 and the CFD General Notation System (CGNS). Input is done either by user specification or with CGNS files. UCLBS is parallelized with the Message Passing Interface (MPI) through the use of Amrex. Geometry is handled internally by reading in an .stl file or a .cgns file. The geometry is solved via a Point-in-Polyhedron problem, since LBM mesh penetrates the surface. Surface distance is accounted for as part of boundary conditions in ELBM. Regridding is performed based on user inputs.

In this paper, the LBM chosen for UCLBS, and its modifications, is discussed further in Section II. The computational setup of UCLBS as well as the problems to be analyzed are discussed in Section IV. Results already obtained and result goals for the full paper are shown and discussed in Section V.

II. Lattice Boltzmann Method

Basic LBM solvers are based on discretized version of the Boltzmann Transport Equation, Equation 1, and the Maxwell-Boltzmann Population Distribution Function (PDF), Equation 2. As most LBM flow solvers are based on the original derivations, they cannot simulate flow much beyond $M \sim 0.2$.

$$\frac{\partial f}{\partial t} + \dot{x} \frac{\partial f}{\partial x} + \dot{p} \frac{\partial f}{\partial p} = \left(\frac{\partial f}{\partial t} \right)_{coll} \quad (1)$$

$$f^{eq} = \frac{\rho}{(2\pi RT)^{3/2}} \exp \left(-\frac{(\xi - u)^2}{2RT} \right) \quad (2)$$

LBM is a stark contrast from more traditional flow solvers which solve the Navier-Stokes (NS) equations. These solvers use equations that solve for the bulk flow and partition it down to a cell size for proper study; solving the NS equations within the cell. LBM starts from bulk statistical averages of molecular movement and scales up the mesoscopic scale to solve the flow at a similar cell or node size as the NS solver [30].

Several elements are common through all LBM models, collision, streaming, and flow calculation. For many, the collision step is based on the Bhatnagar-Gross-Krook (BGK) collision operator [1], Equation 3, the simplest solution to the Boltzmann Transport Equation. After the collision step occurs, the PDF values for the chosen LBM model move to new locations on the chosen lattice. The values for density, velocity, and possibly temperature are then calculated and another collision occurs. Thus, the work flow is collision $\rightarrow$ streaming $\rightarrow$ calculation.

$$\Omega(f) = \frac{1}{\tau^*} (f - f^{eq}) \quad (3)$$

The second common theme through all is the lattice itself. While many lattices exist, all must be isotropic or the physics will be violated. Figure 1 shows three different types. The naming convention entails the number of spatial dimensions followed by the number of directions. Thus, a D2Q9 lattice, as in Figure 1a, has two dimensions and nine directions. Almost all lattices include a rest direction, the node at the center of the lattice, where the post-collision value replaces the distribution function value of the previous timestep. Figure 1b shows a D3Q19 lattice, a common lattice for low-speed simulations.

The post collision values are calculated at a node, and moved along the lattice directions to adjacent nodes. Larger lattices allow further movement, such as the D3Q39 lattice in Figure 1c, which reaches up to three lattice lengths away. These larger lattices allow a wider velocity range at the cost of more computational time and memory. Valid lattices can be found through an analysis of the energy shells and moments of the equations [26].

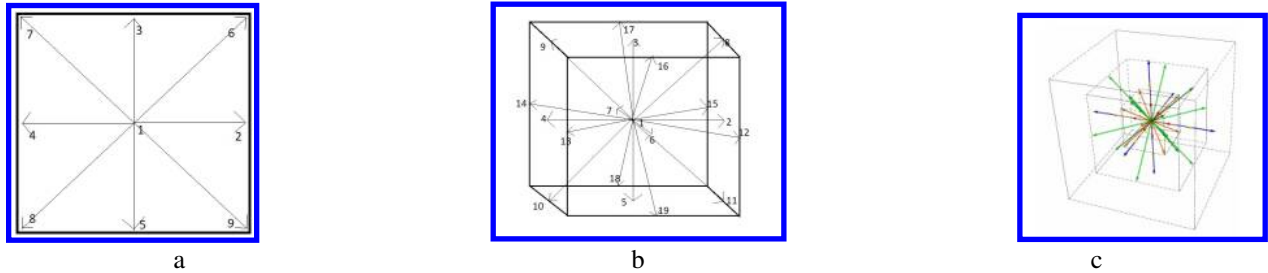


Figure 1 A D2Q9 lattice (a), a D3Q19 lattice (b), and a D3Q39 lattice (c) [3]

Lastly, LBM flow solvers use the same definition for velocity and density, shown in Equation 4. As LBM was originally derived in phase space, the distribution function contains information for both position and velocity. The position information effectively includes the mass at that location as number density of molecules can be used to find the local density.

$$\rho(x, t) = \sum_i f_i(x, t) \quad \rho u_i(x, t) = \sum_i c_i f_i(x, t) \quad (4)$$

In the case of flow solvers using a separate distribution function for energy, Equation 5 is included in the flow variable computations. Those including temperature in a single PDF utilize Equation 6.

$$2\rho C_v T + \rho u^2 = \sum_i c_i^2 f_i + \sum_i g_i \quad (5)$$

$$\rho \epsilon = \frac{D}{2} T = \sum_i |c_i - u_i|^2 f_i \quad (6)$$

A. Entropic Lattice Boltzmann Method

The ELBM of Karlin et al. was chosen for UCLBS [4–7, 11–13, 27, 34]. Every LBM model has its own advantages, and ELBM stays closest to the physics of Kinetic Theory from which it was derived. It is also more adaptable than Simulia's model as the lattice can be changed without dramatic changes to the code or formulation. This means a wider range of problems can be run, and the Mach range can be extended further, without deriving new polynomials for the collision equation.

ELBM uses a double-distribution function (DDF) to handle energy in the model. A second distribution function, g_i , is handled very similarly to the PDF f_i , and is propagated on the same lattice. The function, g_i , contains information on the translational, vibrational, and rotational energy of the molecules it represents. It allows the model to use a variable ratio of specific heats, γ , and variable Prandtl number.

Equation 7 is the discretized solution to Equation 1 in the ELBM framework. The left-hand side represent the PDF at the next timestep after streaming. On the right-hand side, the first term is the PDF at the current timestep, the second term is quasi-equilibrium modification, and the last term is the equilibrium modification. Equation 8 is the equilibrium PDF for momentum and energy, represented in terms of Lagrange multipliers. Equation 9 is the quasi-equilibrium PDF. The equilibrium PDF, f_i^{eq} , is discussed later in Section III.

$$f_i(x + c_i \Delta t, t + \Delta t) = f_i(x, t) + \alpha \beta_1 (f_i^* - f_i) + 2\beta_2 (f_i^{eq} - f_i^*) \quad (7)$$

$$f_i^{eq} = \rho w_i \exp(\chi + c_{i\alpha} \zeta_\alpha + c_i^2 \lambda) \quad g_i^{eq} = (2C_v - D) T f_i^{eq} \quad (8)$$

$$f_i^* = f_i^{eq} + \frac{(\bar{Q}_{\alpha\beta\gamma} - Q_{\alpha\beta\gamma}^{eq})(c_{i\alpha} c_{i\beta} c_{i\gamma} - 3c_{i\gamma} T \delta_{\alpha\beta})}{6T^3} \quad \text{where } \bar{Q}_{\alpha\beta\gamma} = \sum_i f_i (c_{i\alpha} - u_\alpha)(c_{i\beta} - u_\beta)(c_{i\gamma} - u_\gamma) \quad (9)$$

$$\bar{Q}_{\alpha\beta\gamma} = \sum_{i=1}^n (c_{i\alpha} - u_\alpha)(c_{i\beta} - u_\beta)(c_{i\gamma} - u_\gamma) f_i \quad (10)$$

The solution to the Lagrange multipliers from Equation 8 is shown in Equation 11. It is solved iteratively with a multivariate Newton-Raphson iteration, with the Jacobian being shown in Equation 12. Solving for the equilibrium PDF is likely the most computationally expensive part of a single timestep.

$$\begin{bmatrix} \chi_{n+1} \\ \zeta_{\alpha,n+1} \\ \zeta_{\beta,n+1} \\ \zeta_{\gamma,n+1} \\ \lambda_{n+1} \end{bmatrix} = \begin{bmatrix} \chi_n \\ \zeta_{\alpha,n} \\ \zeta_{\beta,n} \\ \zeta_{\gamma,n} \\ \lambda_n \end{bmatrix} - J^{-1} r(\chi, \zeta_\alpha, \lambda) \quad (11)$$

$$J = \begin{bmatrix} \sum \frac{\delta f_i^{eq}}{\delta \chi} & \sum c_{ix} \frac{\delta f_i^{eq}}{\delta \zeta_x} & \sum c_{iy} \frac{\delta f_i^{eq}}{\delta \zeta_y} & \sum c_{iz} \frac{\delta f_i^{eq}}{\delta \zeta_z} & \sum c_i^2 \frac{\delta f_i^{eq}}{\delta \lambda} \\ \sum c_{ix} \frac{\delta f_i^{eq}}{\delta \zeta_x} & \ddots & & & \vdots \\ \sum c_{iy} \frac{\delta f_i^{eq}}{\delta \zeta_y} & & \ddots & & \vdots \\ \sum c_{iz} \frac{\delta f_i^{eq}}{\delta \zeta_z} & & & \ddots & \vdots \\ \sum c_i^2 \frac{\delta f_i^{eq}}{\delta \lambda} & \dots & \dots & \dots & \sum c_i^4 \frac{\delta f_i^{eq}}{\delta \lambda} \end{bmatrix} \quad (12)$$

The energy PDF works similarly to the basic PDF with collision and streaming operations on the same lattice. Equation 13 is the collision equation for energy. Equation 14 defines the relationship between f_i^{eq} and g_i^{eq} as well as the quasi-equilibrium equations and contracted heat flux.

$$g_i(x + c_i \Delta t, t + \Delta t) = g_i(x, t) + \alpha \beta_1 (g_i^* - g_i) + 2\beta_2 (g_i^{eq} - g_i^*) \quad (13)$$

$$g_i^{eq} = (2C_v - D)T f_i^{eq} \quad g_i^* = g_i^{eq} + \frac{(\bar{q}^g - q^{eq}) \cdot c_i}{T} \quad \bar{q}^{eq} = \sum_i g_i(c_i - u) \quad (14)$$

$$\bar{q}^g = \sum_{i=1}^n (c_{i\alpha} - u_\alpha) g_i \quad (15)$$

The entropic relaxation parameter, α , is found by using Boltzmann's H-theorem, Equation 16. By the Second Law of Thermodynamics, the net change in entropy is always positive, thus the roots of $\mathcal{H}$ are always positive. By solving Equation 17 for α , with the restrictions in Equation 18, it is possible to find the local entropic estimate. This estimate acts similarly to a sub-grid scale model in an LES model.

$$\mathcal{H}(f) = \sum_i f_i \ln(f_i) \quad (16)$$

$$\mathcal{H}(f + \alpha(f^{eq} - f)) = \mathcal{H}(f) \quad \text{and let } f_\alpha = f + \alpha(f^{eq} - f) \quad \text{and } \delta_i = \frac{f_i}{f_i - f_i^{eq}} \quad (17)$$

$$\begin{cases} \alpha_{min} &= \max(\delta_i, i \in 1, \dots, n, \quad \text{for } \delta_i < 0) \\ \alpha_{max} &= \min(\delta_i, i \in 1, \dots, n, \quad \text{for } \delta_i > 0) \end{cases} \quad (18)$$

Equations 16-18 are solved via Newton-Raphson iteration. Equation ?? is the resulting equation. In cases where $f_i \sim f_i^{eq}$, the value of α should be set to the BGK value of 2.0, as δ_i approaches ∞ . In cases where $\alpha_{max} < 2$, α should be set directly as α_{max} , as the calculations will be driven to that. The local entropic estimate, α , is discussed further in Section III.

Several parameters are defined to account for local relaxation, Equation 19. These can also be used to extract flow values which do not appear from other equations as readily.

$$\beta_1 = \frac{1}{\frac{2\mu}{\rho T} + 1} \quad \beta_2 = \frac{1}{\frac{2\kappa}{\rho(C_v+1)T} + 1} \quad (19)$$

B. Boundary Conditions

Boundary conditions for ELBM are relatively different from most NS solvers. As the lattice directions will point outside the fluid domain, there will be nothing to fill them in. This is, of course, the job of the boundary conditions. The most straightforward are farfield conditions where the boundary feeds the freestream conditions back into the domain.

Symmetry conditions are more difficult as the distance of the direction to the boundary must be accounted for. In simplest terms, the boundary will feed back the PDF value of a node interacting with it, but it may not be the same node. For example, in the case of a diagonal direction, nodes nearby may provide the information as a psuedo-node which would be on the other side of the symmetry condition.

Most challenging is the non-slip wall. The information must found from the node, the distance to the wall, and the rest of the lattice directions at the node. The original method of doing this was a simple bounceback condition, where the PDF value going into the wall was reversed and placed as the missing PDF value. This, however, is non-physical and only works well in the case where the distance between the node and the wall is exactly half a lattice length.

This leads to the ELBM bounceback method discussed here. Equation 20 shows the method to find the target values for density and any other fluid property needed.

$$\rho_{tgt} = \sum_{i=wall} f_i^{bb} + \sum_{i=fluid} f_i \quad q_{tgt} = \frac{1}{n_{wall}} \sum_{i=wall} \frac{x_{wall}/l_{latt} q_{f,i} + q_{wall,i}}{x_{wall}/l_{latt} + 1} \quad (20)$$

The missing values are found as a function of these target values, Equations. 21&22. Included in those missing values is the non-equilibrium value and nearby gradients.

$$f_i^{miss} = f_i^{eq}(\rho_{tgt}, u_{\alpha,tgt}, T_{tgt}) + f_i^{(1)}(\rho_{tgt}, u_{\alpha,tgt}, T_{tgt}, \nabla u_{tgt}, \nabla T_{tgt}) \quad (21)$$

$$g_i^{miss} = g_i^{eq}(\rho_{tgt}, u_{\alpha,tgt}, T_{tgt}) + g_i^{(1)}(\rho_{tgt}, u_{\alpha,tgt}, T_{tgt}, \nabla u_{tgt}, \nabla T_{tgt}) \quad (22)$$

The non-equilibrium momentum PDF values are found with Equation 23. Equations 24&25 represent the pressure and heat flux gradients.

$$f_i^{(1)} = w_i \left(\frac{P_{\alpha\beta}^{(1)}}{2T^2} (c_{i\alpha}c_{i\beta} - T\delta_{\alpha\beta}) + \frac{Q_{\alpha\beta\gamma}^{(1)}}{6T^3} (c_{i\alpha}c_{i\beta}c_{i\gamma} - 3c_{i\gamma}T\delta_{\alpha\beta}) \right) \quad (23)$$

$$P_{\alpha\beta}^{(1)} = -\frac{\rho T}{2\beta_1} \left(S_{\alpha\beta} - \frac{\partial_\gamma u_\gamma \delta_{\alpha\beta}}{c_v} \right) S_{\alpha\beta} = \frac{\partial u_\alpha}{\partial x_\beta} + \frac{\partial u_\beta}{\partial x_\alpha} \quad (24)$$

$$Q_{\alpha\beta\gamma}^{(1)} = -\frac{\rho T}{2\beta_2} (\partial_\alpha T \delta_{\beta\gamma} + \partial_\beta T \delta_{\alpha\gamma} + \partial_\gamma T \delta_{\alpha\beta}) + u_\alpha P_{\beta\gamma}^{(1)} + u_\beta P_{\alpha\gamma}^{(1)} + u_\gamma P_{\alpha\beta}^{(1)} \quad (25)$$

The energy PDF values, Equation 26, are slightly simpler to find, as several terms cancel out. Equations 27&28 represent the higher order moments near the wall. These are normally part of other equations, and they only show up here because of the non-equilibrium nature of the problem at hand.

$$g_i^{(1)} = w_i \left(K^{(1)} + c_{i\alpha} \frac{q_\alpha^{g(1)}}{T} + (c_{i\alpha}c_{i\beta} - T\delta_{\alpha\beta}) \frac{R_{\alpha\beta}^{(1)}}{2T^2} \right) \quad (26)$$

$$K^{(1)} = -\frac{\rho T (2C_v - D)}{2\beta_1} \left(\frac{\partial_\gamma u_\gamma}{C_v} \right) q_\alpha^{g(1)} = -\frac{\rho T (2C_v - D)}{2\beta_2} \partial_\alpha T + u_\alpha K^{(1)} \quad (27)$$

$$R_{\alpha\beta}^{g(1)} = -\frac{\rho T (2C_v - D)}{2\beta_1} \left(TS_{\alpha\beta} + u_\alpha \partial_\beta T + u_\beta \partial_\alpha T - \frac{\partial_\gamma u_\gamma}{C_v} (2T\delta_{\alpha\beta} + u_\alpha u_\beta) \right) \quad (28)$$

Because of the truncated nature of the lattice length, an adjustment to the local numerical derivatives is required, Equation 29. These derivatives work where x_{wall} is the distance in local lattice lengths to the wall.

$$\frac{\partial q}{\partial x} = \frac{q_{j+1} - q_{wall}}{1 + |x_{wall}|} \text{ or } \frac{\partial q}{\partial x} = \frac{q_{wall} - q_{j-1}}{1 + |x_{wall}|} \quad (29)$$

III. The University of Cincinnati Lattice Boltzmann Solver

Many variations of ELBM exist for compressible flow and higher order thermal flow. The ELBM of Karlin et al. most often uses lattices of the type DnQxⁿ, where x is almost always an odd number greater than 1. While extremely accurate, these lattices can encompass a huge number of nodes, 343 for a D3Q7³ lattice, required for reaching M~2.0. The memory requirements for simulation of something of this nature can be enormous as every node must store 343 f_i values and 343 f_i^{out} values with the same number for the energy PDF values.

The works by Frapolli, show a way forward with a reduced lattice [7]. The new lattice is a D3Q39 lattice with a maximum lattice distance of three lattice lengths. In order to properly account for the reduced lattice, equilibrium PDF calculation must be modified to account for pressure, shear, and heat flux. Equations 30&31 shows this modified equilibrium PDF. Equation 32 shows the pressure and contracted first order heat flux tensor.

$$[\rho, \rho u_\alpha, P_{\alpha\beta}^{eq}, q_\alpha^{eq}] = \sum_i (1, c_i, c_i c_j, c_i c^2) f_i^{eq}(\rho, u, T) \quad (30)$$

$$f_i^{eq} = \rho * \exp \left(\chi + c_{i\alpha} \zeta_\alpha + c_{i\alpha} c_{i\beta} \pi_{\alpha\beta} + c_{i\alpha} c_i^2 \lambda_\alpha \right) \quad (31)$$

$$P_{\alpha\beta}^{eq} = \sum_i c_{i\alpha} c_{i\beta} f_i^{eq} = \rho T \delta_{\alpha\beta} + \rho u_\alpha u_\beta, \quad q_\alpha^{eq} = \sum_i c_{i\alpha} c_i^2 f_i^{eq} = 2\rho u_\alpha \left(T \left(\frac{D}{2} + 1 \right) + \frac{u^2}{2} \right) + \rho u_\alpha u_\beta \quad (32)$$

Solving these equations requires solving a 13x13 Jacobian for three-dimensional flows, similar to Equations 11&12. Thankfully, the Jacobian is symmetric and many terms are repeated. The result is that only around 50 of the 169 terms requires full computation. UCLBS uses an LU decomposition to solve the matrix.

Equations 7-14 normally include a weighting function. Entropic guided equilibrium effectively includes the weighting parameters as part of the equilibrium PDF. Simulations are generally initialized from equilibrium PDF values, thus this weighting is included in the following steps and can be omitted from the collision calculations.

A. Deviation from Standard ELBM

Several small deviation from the standard ELBM form were used to work around issues or accelerate computational speed. The two major deviations are a modified α equation and a pruned three-dimensional lattice.

The modified α equations, Equations 33-36, come from the work of Atif [39]. The approach was changed because very small variations in the distribution function tended to cause the α value to drive toward zero. The implementation of the solution for α prevents this while also being computationally faster.

$$g(\alpha) = -\alpha^2 \beta^2 Q_1 + \alpha Q_2 - Q_3 \text{ where } \beta = \frac{\Delta t}{2\tau + \Delta t} \text{ and } \chi = \frac{f_i^{eq} - f_i}{f_i} \quad (33)$$

$$Q_1 = \sum_{i=1}^n f_i \left(\frac{\chi_i^3}{6} - \frac{h\beta\chi_i^4}{12} + \frac{h^2\beta^2\chi_i^5}{20} - \frac{h^3\beta^3\chi_i^6}{5} \right) \quad (34)$$

$$Q_2 = \sum_{i=1}^n f_i \left(\frac{\chi_i^2}{2} - \frac{2k\beta^2\chi_i^3}{15} \left(\frac{2}{4+k\chi_i} + \frac{1}{4+2k\chi_i} + \frac{2}{4+3k\chi_i} \right) \right) \quad (35)$$

$$Q_3 = \sum_{i=1}^n f_i \left(\frac{60\chi_i^2 + 60\chi_i^3 + 11\chi_i^4}{60 + 90\chi_i + 36\chi_i^2 + 3\chi_i^3} \right) \quad (36)$$

The second major change is the pruned lattice as discussed in Frapolli [7]. This resulted in a D3Q39 lattice that was capable of supersonic flow while losing minimal accuracy. This was verified by Shan [26] including a method to find more possible pruned lattices. The other major effect of pruning the lattice was the removal of weights. This was replaced by a system representing Grad's 13 Moments. Equations 37&38 are the updated representation of the equilibrium distribution function. These equations are solved via multivariate Newton-Raphson method and involve a 13x13 symmetric Jacobian, similar to Equations 11&12.

$$\delta\mathcal{H} + \delta \left(\chi\rho + \zeta_\alpha (\rho u_\alpha) + \pi_{\alpha\beta} P_{\alpha\beta}^{eq} + \lambda_\alpha q_\alpha^{eq} \right) = 0 \quad (37)$$

$$f_i^{eq} = \rho * \exp \left(\chi\rho + \zeta_\alpha c_{i\alpha} + \pi_{\alpha\beta} c_{i\alpha} c_{i\beta} + \lambda_\alpha c_{i\alpha} c_i^2 \right) \quad (38)$$

B. Shifted Lattice

As well as the pruned lattice, UCLBS uses a shifted lattice, when appropriate, for transonic and supersonic simulations. The shifted lattice will be used when the a Mach number, specified in the input file, exceeds M1.0. This requires enough *a priori* knowledge to roughly determine peak Mach numbers in the simulation.

Shifting the lattice moves the rest node upstream, with respect to the primary flow direction, in order to better handle the distribution functions. This shift essentially changes the frame of reference from one at rest, to one moving at M1.0. This is shown in Figure 2, where an unshifted D2Q9 lattice (a) is shown next to a shifted lattice (b).

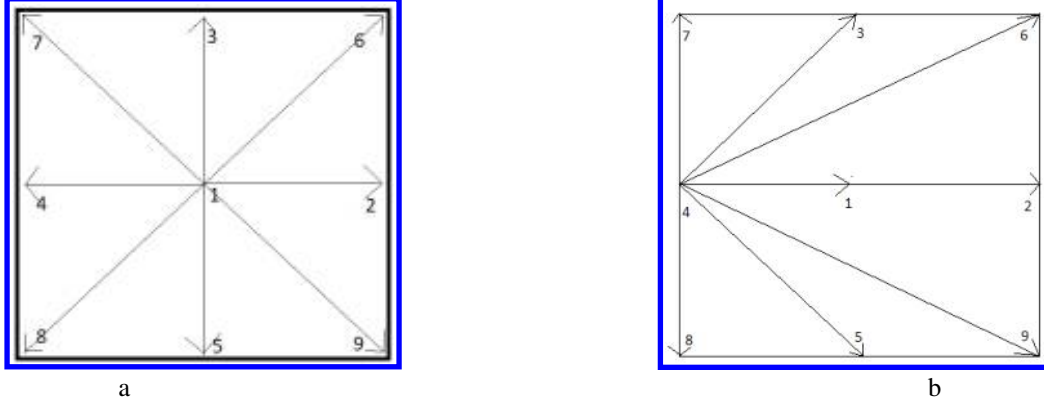


Figure 2 Unshifted (a) and shifted (b) D2Q9 lattices

The system becomes less stable near M1.5 without the shifted lattice because the distribution function or functions pointing against the predominant flow become increasingly small and other parts of the method can exceed these values. This means small perturbations can cause non-physical problems, such as negative f_i and g_i values. The shifted lattice allows for Mach numbers up to ~ 2.5 while still allowing boundary layers to form on non-moving bodies. Unfortunately, the lattice cannot be shifted further if bodies at rest are in the flow as the equations will not converge.

C. Multi-grid for UCLBS

The original timestep workflow for UCLBS was slightly modified to work within the confines of Amrex's systems. The grid refinement algorithm described by Dorschner et al. [4] is as follows:

- 1) Full work flow cycle on the fine grid.
- 2) Streaming step performed on both coarse and fine grid levels.
- 3) Rescale populations at coarse and fine edge nodes.
- 4) Reconstruct populations on two layers of fine edge nodes.
- 5) Perform collision on all grids.

This works with Amrex's multigrid system, but UCLBS must account for the extant but inactive coarse nodes away from the edges. Amrex uses a node-based multigrid (MG) mesh that has complete meshes at every level and data are transferred from the fine grids onto the coarse grids. The region of information exchange between coarse and fine grid consists of a region of ghost nodes on the fine grid level twice the longest lattice direction, and a region in the coarse grid that is equal to the longest lattice direction. In practice, this is six nodes for the fine grid and three nodes for the coarse grid, though they are the same overall distance.

Equation 39 describes the refinement ratio for convective matching. In this system, the grid size is proportional to the timestep, regardless of grid size. For most LBM systems, the chosen refinement ratio is 2. Practically, for a system with two levels, the finer grid level must take two timesteps for every timestep on the coarser level. In simulations with a large number of levels, the finest level will take many more timesteps than the coarsest.

$$r = \frac{dx_c}{dx_f} \quad r = \frac{dt_c}{dt_f} \quad (39)$$

At the interface between the grids, some interpolation must be performed to ensure continuity across the boundary. For this, the distribution function, the equilibrium distribution function, and a non-equilibrium distribution function are used to perform the interpolation calculations. Equation 40 shows the distribution function relationships and how they relate to each other across levels. Equation 41 is the viscosity ratio between the levels. Notably, the viscosity change modifies the β_1 relaxation parameter so it is not the same across the interface.

$$f_i = f_i^{eq} + f_i^{neq} \quad \bar{f}_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}} \bar{f}_{i,c}^{neq} \quad \bar{g}_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}} \bar{g}_{i,c}^{neq} \quad (40)$$

$$\nu_f = r\nu_c \quad (41)$$

Equation 42 defines the relation between f_i^{neq} and $\bar{f}_i^{neq}$ with this relation being true regardless of level. Equation 44 defines the local pressure and heat flux. When Equation 40 is put into Equation 43 and Equation 42 is used at the coarse level, the result is Equation 45. While β_2 is the same across the grid levels as the heat capacity does not change, β_1 must change to maintain a continuous lattice Reynolds number.

$$\bar{f}_i^{neq} = f_i^{neq} + \frac{c_{i\alpha}c_{i\beta}c_{i\gamma} - 3c_{i\gamma}T\delta_{\alpha\beta}}{6T^3} \left(u_\alpha P_{\beta\gamma,c}^{(1)} + u_\beta P_{\alpha\gamma,c}^{(1)} + u_\gamma P_{\alpha\beta,c}^{(1)} - Q_{\alpha\beta\gamma,c}^{(1)} \right) \quad (42)$$

$$f_{i,f}^{neq} = \bar{f}_{i,f}^{neq} - \frac{\beta_{2,c}}{r\beta_{2,f}} \frac{c_{i\alpha}c_{i\beta}c_{i\gamma} - 3c_{i\gamma}T\delta_{\alpha\beta}}{6T^3} \left(u_\alpha P_{\beta\gamma,c}^{(1)} + u_\beta P_{\alpha\gamma,c}^{(1)} + u_\gamma P_{\alpha\beta,c}^{(1)} - Q_{\alpha\beta\gamma,c}^{(1)} \right) \quad (43)$$

$$P_{\alpha\beta}^{(1)} = \sum_i c_{i\alpha}c_{i\beta} (f_i - f_i^{eq}) \quad Q_{\alpha\beta\gamma}^{(1)} = \sum_i c_{i\alpha}c_{i\beta}c_{i\gamma} (f_i - f_i^{eq}) \quad (44)$$

$$f_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}} \left(f_{i,c}^{neq} \right) + \left(\frac{\beta_{1,c}}{r\beta_{1,f}} - \frac{\beta_{2,c}}{r\beta_{2,f}} \right) \frac{c_{i\alpha}c_{i\beta}c_{i\gamma} - 3c_{i\gamma}T\delta_{\alpha\beta}}{6T^3} \left(u_\alpha P_{\beta\gamma,c}^{(1)} + u_\beta P_{\alpha\gamma,c}^{(1)} + u_\gamma P_{\alpha\beta,c}^{(1)} - Q_{\alpha\beta\gamma,c}^{(1)} \right) \quad (45)$$

For the energy distribution function, the equations are quite similar, as shown in Equations 46&47. The interpolation to the fine grid is shown in Equation 48. While the same pattern can be followed as the distribution function, Equation 49 shows the final energy distribution function in a condensed form.

$$\bar{g}_i^{neq} = g_i^{neq} + \frac{c_{i\alpha} \left(u_\alpha K^{(1)} - q_\alpha^{(1)} \right)}{T} \quad (46)$$

$$K^{(1)} = \sum_i (g_i - g_i^{eq}) \quad q_\alpha^{(1)} = \sum_i c_{i\alpha} (g_i - g_i^{eq}) \quad (47)$$

$$g_{i,f}^{neq} = \bar{g}_{i,f}^{neq} + \frac{\beta_{2,c}}{r\beta_{2,f}} \frac{c_{i\alpha} \left(u_\alpha K_c^{(1)} - q_{\alpha,c}^{(1)} \right)}{T} \quad (48)$$

$$g_{i,f}^{neq} = \frac{\beta_{1,c}}{r\beta_{1,f}} g_{i,c}^{neq} + \left(\frac{\beta_{1,c}}{r\beta_{1,f}} + \frac{\beta_{2,c}}{r\beta_{2,f}} \right) \frac{c_{i\alpha} \left(u_\alpha K_c^{(1)} - q_{\alpha,c}^{(1)} \right)}{T} \quad (49)$$

With regards to the g_i^{neq} , the same information can be obtained from Equation 9 as g_i^{eq} is a function of f_i^{eq} . Since the Lagrange multipliers are known and interpolated at both levels, use of this calculation method is likely far simpler and more straightforward. Equation 9 also helps to keep the information internally consistent with other parts of the ELBM system.

IV. Computational Setup

Multiple setup and test cases have been created to analyze whether UCLBS is correctly simulating the fluid flow. UCLBS was written in Fortran 95 for a mixture of speed and accessibility.

UCLBS is a MG flow solver which uses Amrex [2, 3] for its AMR features. Amrex is a robust multigrid code with one-, two-, and three-dimensional functionality for both cells and nodes. Built in functions allow the filling of ghost nodes or cells, averaging the finer grid levels onto the coarser ones, and many other, highly adaptable features. Of primary interest for this work is the inclusion of MPI software within it. The mesh is automatically divided and assigned to processors. Functions built into Amrex allow passing of information between the boxes and levels.

Output is done using CGNS [35] and Plot3D format files. While Amrex has its own output methods, CGNS is far more universal to flow solvers and visualization tools. CGNS allows for parallel output through HDF5 and MPI. Issues were found with output and CGNS functionality, so Plot3D output was used to simplify output until CGNS functionality could be restored.

A. Multigrid Setup and Methodology

As previously mentioned, all multigrid functionality in UCLBS is handled by Amrex. The user of UCLBS is able to control the how often Amrex adapts the mesh to the new flow conditions. The adaptation is done through finding the gradient magnitude of a chosen parameter and is described in greater detail in Section IV.B.

LBM grid is almost always rigid and cubic. It is possible to have a non-cubic grid [36], but more calculation are required to maintain isotropy. When MG or MD mesh setups are used with LBM, the refinement ratio between levels needs to be an integer to prevent extra advection steps. The refinement ratio of choice is almost always 2. A factor of two means an increased number of overlapping nodes, which allows data to be shared more easily between levels with fewer computations.

Amrex breaks the grid at every level into multiple boxes. These boxes are distributed such that an MPI process owns the box and its data, and the other processes cannot access it. Propagation from nearby boxes and other grid levels is handled via ghost nodes. Ghost nodes hold data but are not part of the computations. They are only needed for the output distribution functions as the outputs will be swapped with the current distribution functions.

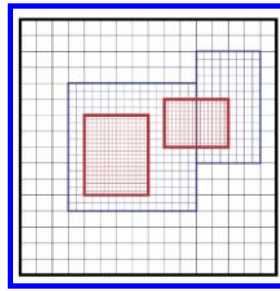


Figure 3 Example of a multigrid setup with Amrex [3]

The initial MG setup is based on user input. A minimum box corner, a maximum box corner, and a grid level are read in from the input file. The grid is then refined down to the user specified level, assuming it is not finer than the specified maximum grid level, before timestepping begins. A nested shell of at least 2 coarse lattice lengths is left at the coarser grid level. This ensures the ability for flow to properly develop on a layer as well as providing sufficient overlap between coarse and fine layers. Within the user specified box, the mesh will never be coarsened as the mesh size at that

The ghost nodes overlap the edges of the existing grid for a level. Ghost nodes do not participate in computations but act as storage for PDFs for interpolation between levels. Filling and interpolation are handled by Amrex as a part of the timestepping procedure. Propagation during the streaming step will occur for all nodes on a level, including ghost nodes. In actual practice of using UCLBS, six ghost nodes must be used on every side because of the D3Q39 lattice. Unfortunately, this is extremely wasteful of memory resources so the largest possible box size must be chosen to maximize performance.

B. Refinement Parameter and Its Application

A refinement parameter was found for use in LBM simulations based on the local rate of change. Within UCLBS, it is called the gradient-gradient magnitude (GGM). It consists of taking the gradient magnitude of a parameter while removing the local cell/node size dependency then performing the task again. This highlights regions of rapid change in a parameter regardless of local grid size. The current basis is somewhat qualitative as the parameter is not scaled to a range. For example, while temperature for a simulation may vary little, the velocity will likely have a range from 0 to at or above the freestream velocity.

The need arose for this parameter while working with another LBM solver. While it was possible to view the results of that solver in great detail, it was difficult to know if the results were sufficiently resolved by the mesh. Observing the gradients was useful, but high gradients will always exist in fluid flows. However, the gradient magnitude of the gradient magnitude applied to the velocity magnitude was able to much better illustrate if the rate of change of the flow values was reasonable. Initial results were more qualitative in nature, as local high points were found in regions of rapid change, such as behind a turbulent trip and in a mixing layer near the trailing edge of the separator. Some empiricism was used, but a value of ~ 0.05 was found to be a threshold for the cells needing more refinement, regardless of refinement level.

Removal of the grid spacing was found beneficial because a region of high gradient may remain high throughout with its inclusion. When removed, the derivatives at a given refinement level become cell or node based and naturally

decreases at finer grid levels.

The idea of GGM is to have a simple system for refinement with no *a priori* knowledge required. Without GGM, determining if a gradient is excessively large for the solver may require knowing that gradient beforehand and approximating an appropriate grid spacing in that region.

Equation 50 is a basic nondimensional value, to be used throughout the equations. Equation 51 is the gradient magnitude (GM) of the value multiplied through with the local lattice length. Equation 52 is the gradient magnitude of the gradient magnitude (GGM), the refinement parameter being explored in this paper.

$$Q^* = \frac{Q}{Q_{ref}} \quad (50)$$

$$GM = \sqrt{\left(\left(\frac{\partial Q^*}{\partial x}\right)^2 + \left(\frac{\partial Q^*}{\partial y}\right)^2 + \left(\frac{\partial Q^*}{\partial z}\right)^2\right)} * l_{latt} \quad (51)$$

$$\|\nabla \cdot \|\nabla \cdot Q^*\| * l_{latt}^2 = GGM = \sqrt{\left(\left(\frac{\partial GM}{\partial x}\right)^2 + \left(\frac{\partial GM}{\partial y}\right)^2 + \left(\frac{\partial GM}{\partial z}\right)^2\right)} * l_{latt} \quad (52)$$

These equations form the backbone of the AMR strategy in UCLBS. During a regrid step with Amrex, the GGM value is computed for all nodes at all relevant levels. Those nodes with values above the local GGM threshold are flagged for refinement. When this information is passed back to Amrex, the software refines the mesh around all flagged nodes. Amrex also redistributes all MPI information in accordance with the new mesh. The post-regrid step determines whether new nodes are fluids or solids and interpolates all PDF and flow variable data.

Figure 4 shows two examples of GGM at work in UCLBS. The example is flow around a cylinder at $Re \sim 300k$ with atmospheric conditions on a single level of refinement. Figure 4a shows the u-velocity with streamlines of flow over the cylinder. Figure 4b shows GGM for the velocity magnitude. The large values near the cylinder help illustrate the rapid velocity change of flow near a wall. The recirculation regions behind the cylinder do not have the high values of GGM because the velocity is comparatively low in the region. Figure 4c shows GGM applied to the x-y fluidic shear stress. The region near the cylinder shows the peak value as the flow moves around it.

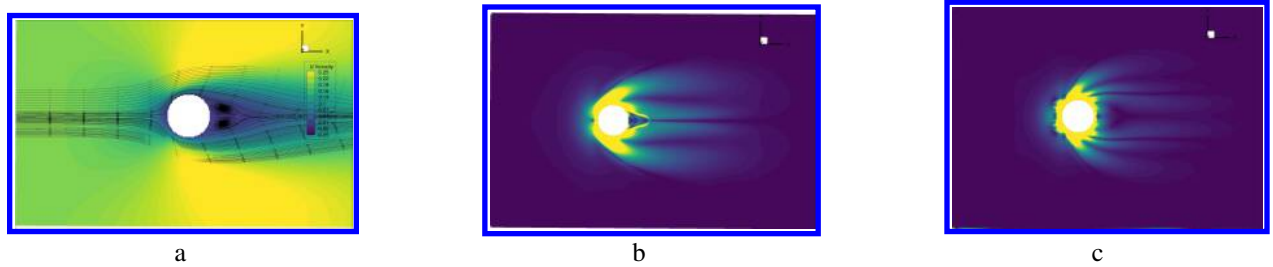
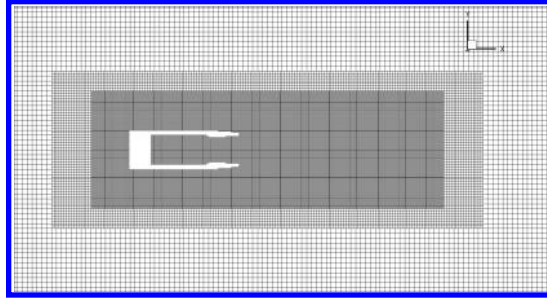


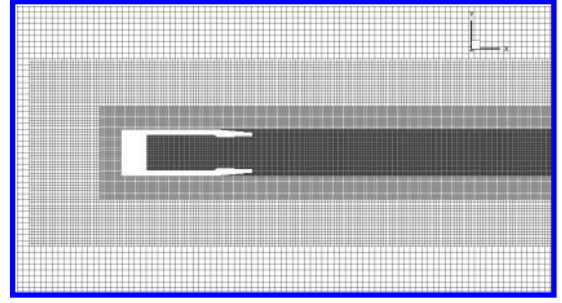
Figure 4 u-velocity (a) and GGM of flow around a cylinder for velocity magnitude (b) and xy-shear (c)

Several caveats exist for the use of GGM. The first is that GGM will be a more expensive computational process than other mesh refinement parameters because of the need to take derivatives twice while temporarily saving the resulting value from the first derivative. The next caveat is that over-refinement is possible if the refinement values are set too permissively initially. A reasonable limit should be set on the number of levels, the allowable refinement region, and the parameter value to flag nodes for refinement.

Figure 5 shows the initial mesh (a) and final mesh (b) on a plane through the centerline of an axisymmetric supersonic nozzle. The refinement decreased the overall size around the nozzle, but also added a finer layer around the nozzle exhaust. An upper limit was placed on the refinement to ensure good computational and memory usage.



a



b

Figure 5 Mesh for supersonic nozzle at the start of the simulation (a) and end (b)

V. Current Results

UCLBS results consist of three major section, low-Mach boundary layer results, a supersonic shock generator, and a converging-diverging nozzle. The boundary layer results show the functionality of the code to replicate viscous physics. The shock generator shows that the code allows for high speed flow. Lastly, the converging-diverging nozzle shows that the code can produce accurate results with large velocity differences.

A. Low Mach Results

Several cases are available for study for this work, but the basic cases of a flat plate and a pipe are used to demonstrate the flow physics. The flow conditions are carefully chosen to ensure laminar or turbulent flow, as there is no turbulence model in UCLBS. Instead, the ELBM model acts as a sub-grid scale model from large-eddy simulations (LES) and turbulence is determined by the flow conditions only.

$$\delta_{lam} \approx \frac{5x}{\sqrt{Re_x}} \quad \delta_{turb} \approx 0.37 * \frac{x}{Re_x^{1/5}} \quad (53)$$

Figure 6 shows a profile of the laminar boundary layer with interpolated points taken at $x = 0.25 \text{ m}$ from the inlet. Actual nodes are indicated in red. The boundary layer height is $\sim 0.011 \text{ m}$. The estimate from Equation 53 calculates the boundary layer height to be $\sim 0.004 \text{ m}$ at the same location. Data were taken from this case at $\sim 0.025 \text{ s}$, while the characteristic time, is 0.1 s . The Reynolds number of the flow over the plate was $100k$.

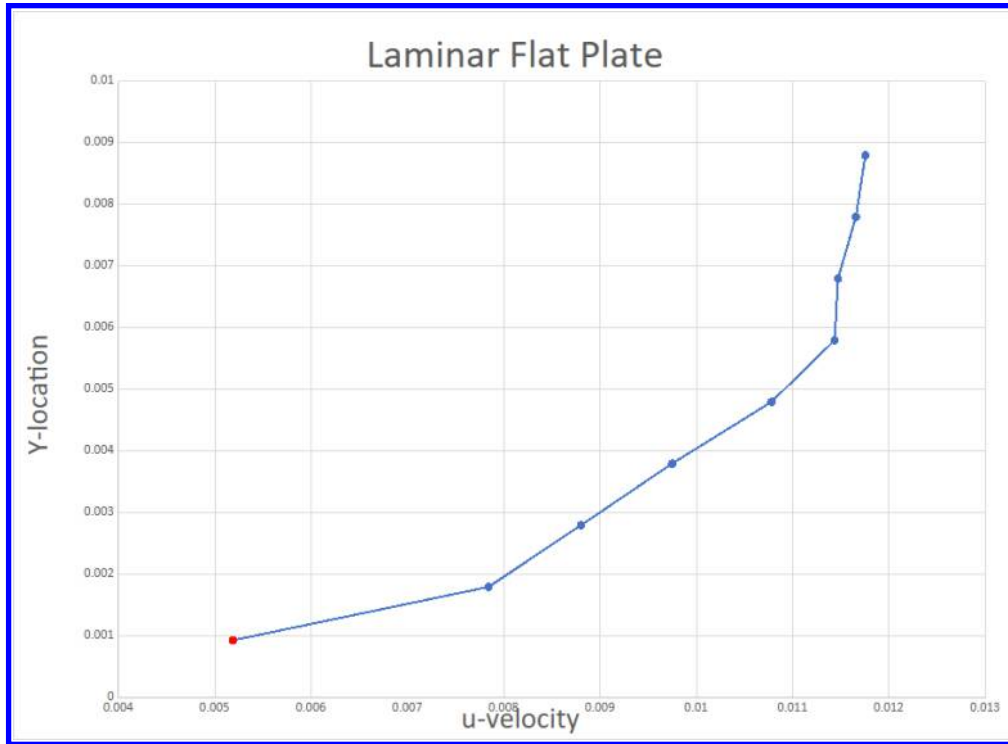


Figure 6 Laminar boundary layer profile

Figure 7 shows the turbulent flat plate boundary layer. The data were taken from a spanwise slice at $x = 0.35$ m from the inlet and time averaged over 16 outputs and with each slice also being averaged across the entire span. The data were taken from ~ 0.00298 - 0.00512 s, and the characteristic time is 0.008 s.

Figure 7 shows the boundary layer profile normal to the wall. The red line is an approximation from Eq. 54, the classic $1/7$ th power law profile. The turbulent boundary layer from UCLBS is not as full as that from the approximation. Eq. 53 estimates the height to be ~ 0.08 m at this location, while the actual height is ~ 0.11 m. This corresponds to the same increased viscosity as the laminar plate, indicating a problem to be found in UCLBS.

$$\bar{u} \approx u_{\infty} \left(\frac{y}{\delta} \right)^{1/7} \quad (54)$$

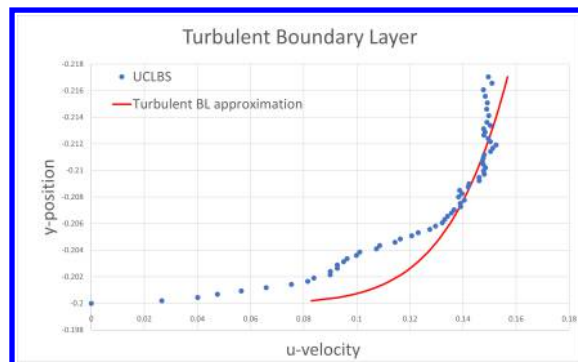


Figure 7 Instantaneous turbulent boundary layer profile

Figure 8 shows a results from a different simulation with the same conditions but a coarser mesh. The profiles are each instantaneous boundary layer profiles with the blue being the first in time, the orange the second, and the gray the last timestep taken. The gray profile was taken when a slight oscillation has started within the boundary layer.

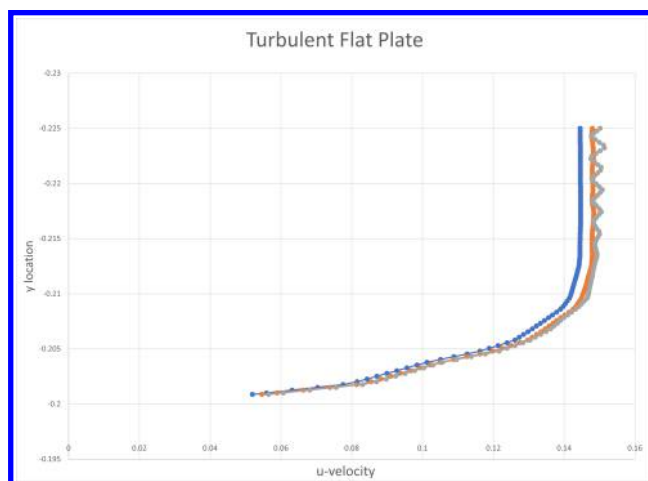


Figure 8 Instantaneous turbulent flat plate

B. Shock Generator

A 10° shock generator is placed in the flow at Mach 1.5. The shifted lattice is used for this simulation as the flow is too close to the Mach limit of the unshifted lattice. As with all work in UCLBS, this is a viscous simulation so the shock will not be straight because of the formation of the boundary layer. Figure 9a shows the final grid resolution for the case. A single, user specified refinement pass was performed in this case, resulting in three total grid levels.

Figure 9b shows a Mach number contour of the center plane of the simulation. A small boundary layer can be seen from the tip of the generator. The slight curvature of the shock can be attributed to the formation of the boundary layer on the surface of the shock generator. Small expansion fans from the effective wall shape of the momentum thickness will tend to cause this in viscous flows. Figure 9c shows the pressure contours for the shock generator.

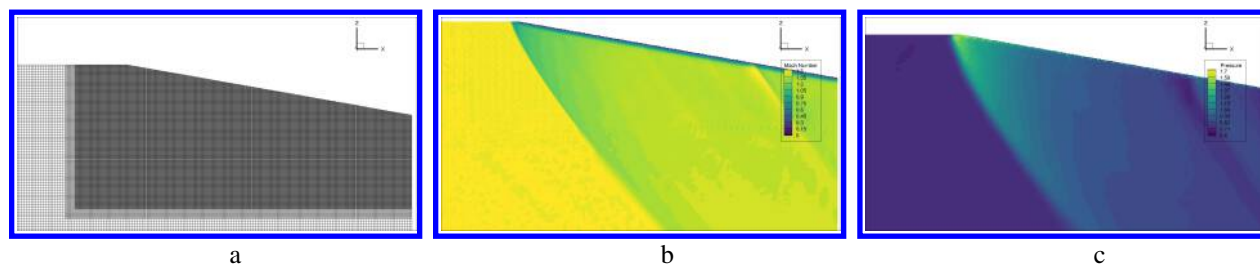


Figure 9 Mach number for 10° shock generator

The initial shock angle and pressure change are in line with the inviscid, analytical values. A small issue with the grid can be seen near the toward the right side of the images. This is likely an issue with the grid solution not correctly identifying adjacent nodes as solid, leading to the small deviation.

C. Converging-Diverging Nozzle

Lastly, a converging-diverging nozzle from Munday et al. [40], with LES from Liu et al. [41] as a verification, is shown to demonstrate the AMR capability and the unsteady capture of the jet exhaust.

As in Section V.B, the shifted lattice is used. The design Mach number of the nozzle is 1.5, but the shape causes peak Mach numbers of 1.6 in the exhaust. This would causes issues with the unshifted lattice.

Figure 10 shows the Mach number of the jet exhausting into the fluid volume. The coarse regrid occurred just before Figure 10a, so it is refined to grid level 3. The other images are as the jet progressing downstream. Figure 10c has the jet exceeding the limits of the fine grid, but the AMR system is able to detect this during the next regrid cycle. The jet is again mostly contained in the fine grid in Figure 10d.

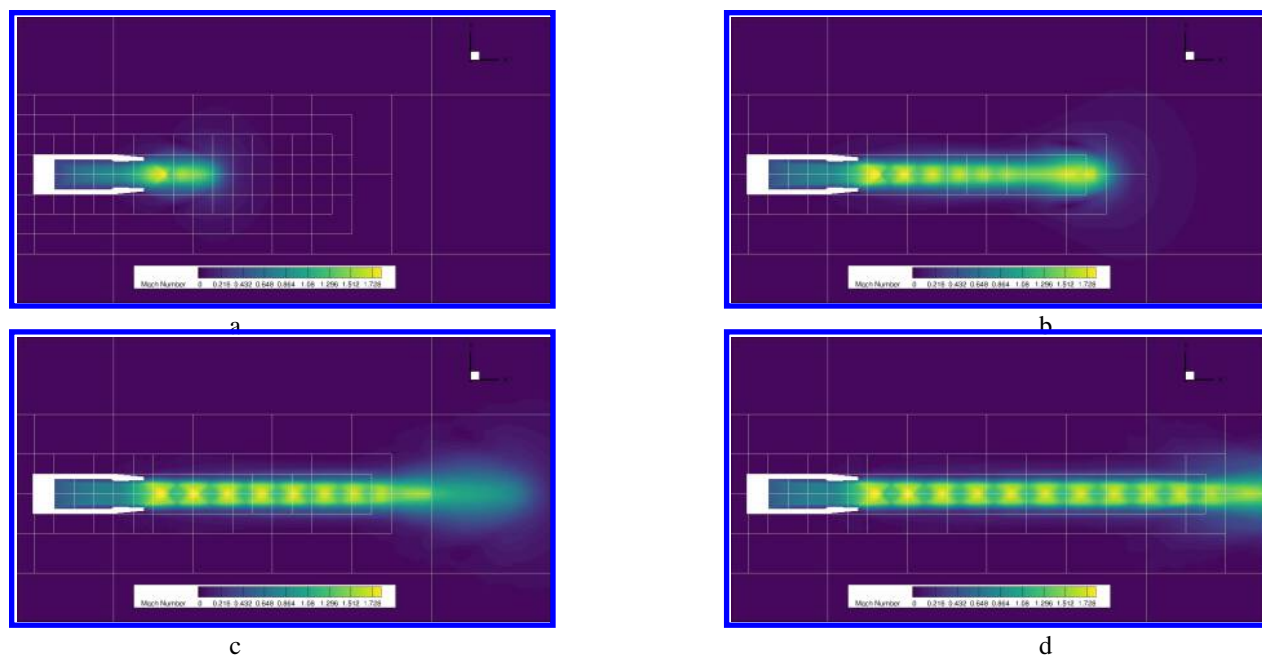


Figure 10 Supersonic jet exhausting into freestream at 0.002 s (a) 0.004 s (b) 0.006 s (c) 0.008 s (d)

Figure 11 shows a comparison of the simulated and experimental jet wakes. The experimental jet wake is averaged over many images, while the simulated wake is instantaneous. It can be seen that the simulated jet wakes compares well to the experiment. The dissipation point, where the wake can be said to end, is similar as well, ~ 12.5 nozzle diameters for the experiment and ~ 14 nozzle diameters for the simulation.

The images are lined up also to the same axis for a more direct comparison. It can be seen the shock cell size is very similar. The experiment dissipates faster, but that image was composed of many images and the image from UCLBS is instantaneous. It can be seen, that the wake is unsteady and wavy. If averaged over many images like the experiment, it will appear more dissipated like the experiment. It is shown as an instantaneous image to illustrate the physical mechanism leading to that dissipation.

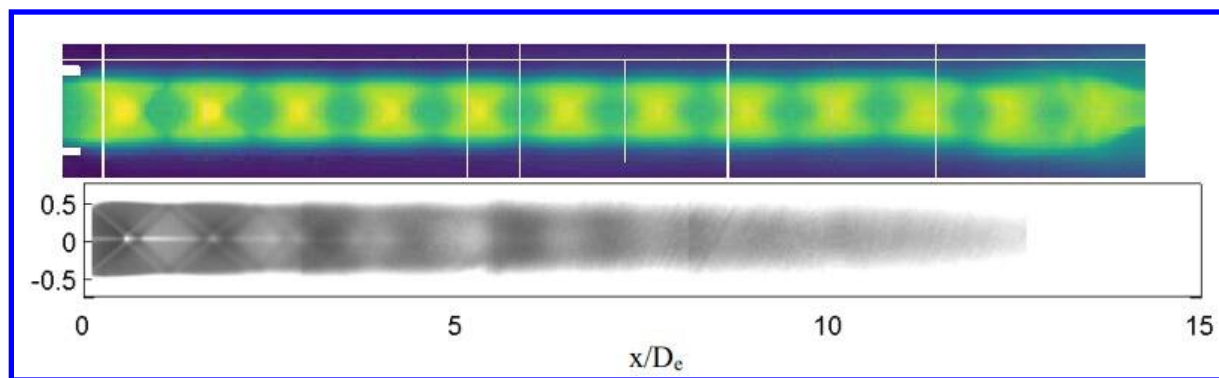


Figure 11 Comparison of UCLBS (above) jet wake to experiment [40] (below)

The results within the nozzle are somewhat different. Figure 12 shows the Mach number through the nozzle throat in UCLBS and compares it to the work of Liu et al. UCLBS does not seem to go supersonic in the cone pattern seen in the LES simulation near the centerline, but it does reproduce similar shock/expansion angles. This is likely due to insufficient local grid resolution near the centerline for the nozzle. Higher resolution studies are left for future work.

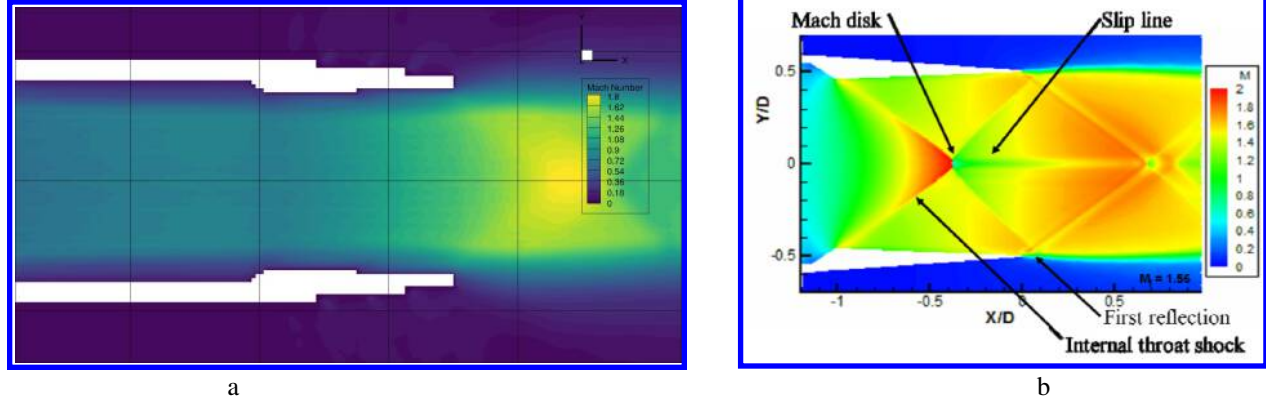


Figure 12 Comparison of Mach number within the throat for UCLBS (a) and Liu et al. [41] (b)

VI. Conclusions

While the results produced so far are good, several small issues prevent UCLBS from being truly accurate as a flow solver. It is desired that others can further UCLBS and build upon the foundation built by this work. Unfortunately, some of the desired projects to improve the code are extremely large, and cannot be completed casually. And, as with all programs, continuous improvement is a goal itself.

The supersonic cases show good agreement with previous work, but the low speed cases can be improved upon. The AMR system shows promise for future work. Overall, UCLBS shows promise, but would benefit from more development.

References

- [1] Bhatnagar, P. L., Gross, E. P., and Krook, M., "A Model for Collision Processes in Gases. I. Small Amplitude Processes in Charged and Neutral One-Component Systems," *Physical Review*, vol. 94, 1954, pp. 511–525. doi:10.1103/physrev.94.511
- [2] Zhang, W., et al., "AMReX: A Framework for Block-Structured Adaptive Mesh Refinement," *Journal of Open Source Software*, vol. 4, 2019, pp. 1370-1373. doi: 10.21105/joss.01370
- [3] "AMReX Documentation", Release 20.07-dev, Block-Structured AMR Co-Design Center, Berkeley, CA, June 2020
- [4] Dorschner, B., Frapolli, N., Chikatamarla, S. S., and Karlin, I. V., "Grid refinement for entropic lattice Boltzmann models," *Physical Review E*, vol. 94, 2016. doi:10.1103/physreve.94.053311
- [5] Frapolli, N., Chikatamarla, S. S., and Karlin, I. V., "Entropic lattice Boltzmann model for compressible flows," *Physical Review E*, vol. 92, 2015. doi: 10.1103/physreve.92.061301
- [6] Frapolli, N., Chikatamarla, S. S., and Karlin, I. V., "Entropic lattice Boltzmann model for gas dynamics: Theory, boundary conditions, and implementation," *Physical Review E*, vol. 93, Jul. 2016. doi: 10.1103/physreve.93.063302
- [7] Frapolli Nicolò, "Entropic lattice Boltzmann models for thermal and compressible flows," dissertation, 2017. doi: 10.3929/ethz-a-010890892
- [8] Shan, X., Yuan, X.-F., and Chen, H., "Kinetic theory representation of hydrodynamics: a way beyond the Navier–Stokes equation," *Journal of Fluid Mechanics*, vol. 550, 2006, p. 413. doi: 10.1017/s0022112005008153
- [9] Kun, Q., "Development of Lattice Boltzmann Method for Compressible Flows," dissertation, 2008.
- [10] Maxwell, J. C., "IV. On the dynamical theory of gases," *Philosophical Transactions of the Royal Society of London*, vol. 157, 1867, pp. 49–88. doi: 10.1098/rstl.1867.0004
- [11] Ansumali, S., and Karlin, I. V., "Stabilization of the lattice Boltzmann method by the H-theorem: A numerical test," *Physical Review E*, vol. 62, Jan. 2000, pp. 7999–8003. doi: 10.1103/physreve.62.7999
- [12] Karlin, I., and Asinari, P., "Factorization symmetry in the lattice Boltzmann method," *Physica A: Statistical Mechanics and its Applications*, vol. 389, 2010, pp. 1530–1548. doi: 10.1016/j.physa.2009.12.032

- [13] Chikatamarla, S. S., and Karlin, I. V., "Lattices for the lattice Boltzmann method," *Physical Review E*, vol. 79, Jan. 2009. doi:10.1103/physreve.79.046701
- [14] Fares, E., Wessels, M., Zhang, R., Sun, C., Gopalaswamy, N., Roberts, P., Hoch, J., and Chen, H., "Validation of a Lattice-Boltzmann Approach for Transonic and Supersonic Flow Simulations," 52nd Aerospace Sciences Meeting, 2014. doi:10.2514/6.2014-0952
- [15] Fares, E., Duda, B., and Khorrami, M. R., "Airframe Noise Prediction of a Full Aircraft in Model and Full Scale Using a Lattice Boltzmann Approach," 22nd AIAA/CEAS Aeroacoustics Conference, 2016. doi:10.2514/6.2016-2707
- [16] Mann, A., Perot, F., Kim, M.-S., Casalino, D., and Fares, E., "Advanced Noise Control Fan Direct Aeroacoustics Predictions Using a Lattice-Boltzmann Method," 18th AIAA/CEAS Aeroacoustics Conference (33rd AIAA Aeroacoustics Conference), Apr. 2012. doi:10.2514/6.2012-2287
- [17] Casalino, D., Ribeiro, A. F., Fares, E., Noelting, S. E., Mann, A., Perot, F., Li, Y., Lew, P.-T., Sun, C., Gopalakrishnan, P., Zhang, R., Chen, H., and Habibi, K., "Towards Lattice-Boltzmann Prediction of Turbofan Engine Noise," 20th AIAA/CEAS Aeroacoustics Conference, 2014. doi:10.2514/6.2014-3101
- [18] Casalino, D., Hazir, A., and Mann, A., "Turbofan Broadband Noise Prediction using the Lattice Boltzmann Method," 22nd AIAA/CEAS Aeroacoustics Conference, 2016. doi:10.2514/6.2016-2945
- [19] Yu, D., Mei, R., and Shyy, W., "A multi-block lattice Boltzmann method for viscous fluid flows," *International Journal for Numerical Methods in Fluids*, vol. 39, 2002, pp. 99–120. doi:10.1002/fld.280
- [20] Guzik, S. M., Weisgraber, T. H., Colella, P., and Alder, B. J., "Interpolation methods and the accuracy of lattice-Boltzmann mesh refinement," *Journal of Computational Physics*, vol. 259, 2014, pp. 461–487. doi:10.1016/j.jcp.2013.11.037
- [21] Dupuis, A., Chopard, B., "Theory and applications of an alternative lattice Boltzmann grid refinement algorithm," *Physical Review E*, vol. 67, 2003, pp.0066707
- [22] Nie, X., Shan, X., and Chen, H., "Lattice-Boltzmann / Finite-Difference Hybrid Simulation of Transonic Flow," 47th AIAA Aerospace Sciences Meeting including The New Horizons Forum and Aerospace Exposition, 2009. doi:10.2514/6.2009-139
- [23] Luo, Y., "Adaptive nearest-nodes finite element method guided by gradient of linear strain energy density," *Finite Elements in Analysis and Design*, vol. 45, 2009, pp. 925–933. doi:10.1016/j.finel.2009.09.004
- [24] Marcuzzi, F., Cecchi, M., and Venturin, M., "An anisotropic unstructured triangular adaptive mesh algorithm based on error and error gradient information," *Mathematics and Computers in Simulation*, vol. 78, 2008, pp. 645–652. doi:10.1016/j.mat-com.2008.04.006
- [25] Barad, M., Kocheemoolayil, J., Stich, G., Kiris, C., "Performance Enhancements for the Lattice-Boltzmann Solver in the LAVA Framework," 10th International Conference on Computational Fluid Dynamics, 2018
- [26] Shan, X., "The mathematical structure of the lattices of the lattice Boltzmann method," *Journal of Computational Science*, vol. 27, 2016, pp. 475–481, <https://doi.org/10.1016/j.jocs.2016.03.002>
- [27] Dorschner, B., Chikatamarla, S.S., Karlin, I.V., "Transitional flows with the entropic lattice Boltzmann method," *Journal of Fluid Mechanics*, vol 824, 2017, pp.388–412, doi:10.1017/jfm.2017.356
- [28] Baeza, A., Martínez-Gavara, A., and Mulet, P., "Adaptation based on interpolation errors for high order mesh refinement methods applied to conservation laws," *Applied Numerical Mathematics*, vol. 62, 2012, pp. 278–296. doi:10.1016/j.apnum.2011.09.008
- [29] Xiao, X., Feng, X., and Li, Z., "A gradient recovery-based adaptive finite element method for convection-diffusion-reaction equations on surfaces," *International Journal for Numerical Methods in Engineering*, vol. 120, Jun. 2019, pp. 901–917. doi:10.1002/nme.6163
- [30] Kruger, T., Kusumaatmaja, H., Kuzmin, A., Shardt, O., Silva, G., Viggien, E.M., (2017). "The Lattice Boltzmann Method". Switzerland: Springer International Publishing
- [31] Wolf-Gladrow, D.A., (2000) *Lattice-Gas Cellular Automata and Lattice Boltzmann Models*. Berlin, Germany: Springer Verlag Berlin Heidelberg
- [32] O'Rourke, J. (1998) *Computational Geometry in C*. New York, NY, USA: Cambridge University Press

- [33] Touil, H., Ricot, D., and L  v  que, E., "Direct and large-eddy simulation of turbulent flows on composite multi-resolution grids by the lattice Boltzmann method," *Journal of Computational Physics*, vol. 256, 2014, pp. 220–233. doi:10.1016/j.jcp.2013.07.037
- [34] Chikatamarla, S. S., and Karlin, I. V., "Lattices for the lattice Boltzmann method," *Physical Review E*, vol. 79, Jan. 2009. doi:10.1103/physreve.79.046701
- [35] Rumsey, C.L., Poirier, D.M.A., Bush, R.H., and Towne, C.E., "CFD General Notation System A User's Guide to CGNS," A User's Guide to CGNS Available: http://cgns.github.io/CGNS_docs_current/user/index.html.
- [36] Wang, Y., Shu, C., and Teo, C., "Thermal lattice Boltzmann flux solver and its application for simulation of incompressible thermal flows," *Computers & Fluids*, vol. 94, 2014, pp. 98–111. doi:10.1016/j.compfluid.2014.02.006
- [37] Li, K., and Zhong, C., "A lattice Boltzmann model for simulation of compressible flows," *International Journal for Numerical Methods in Fluids*, vol. 77, 2014, pp. 334–357. doi:10.1002/fld.3984
- [38] Qiu, R.-F., Zhu, C.-X., Chen, R.-Q., Zhu, J.-F., and You, Y.-C., "A double-distribution-function lattice Boltzmann model for high-speed compressible viscous flows," *Computers & Fluids*, vol. 166, 2018, pp. 24–31. doi: 10.1016/j.compfluid.2018.01.039
- [39] Atif, M., Kolluru, P. K., Thantapally, C., and Ansumali, S., "Essentially Entropic Lattice Boltzmann Model," *Physical Review Letters*, vol. 119, 2017. doi:10.1103/physrevlett.119.240602
- [40] Mundy, D., Gutmark, E., Liu, J., and Kailasanath, K., "Flow Structure of Supersonic Jets from Conical C-D Nozzles," 39th AIAA Fluid Dynamics Conference, June 2009.
- [41] Liu, J., Kailasanath, K., Ramamurti, R., Munday, D., Gutmark, E., Lohner, R., "Large-Eddy Simulation of a Supersonic Jet and Its Near-Field Acoustic Properties - Methodology & Validation," 47th Aerospace Sciences Meeting, January 2009.
- [42] Li, W., and Luo, L.-S., "An implicit block LU-SGS finite-volume lattice-Boltzmann scheme for steady flows on arbitrary unstructured meshes," *Journal of Computational Physics*, vol. 327, 2016, pp. 503–518. doi:10.1016/j.jcp.2016.09.038
- [43] Seta, T., Rojas, R., Hayashi, K., and Tomiyama, A., "Implicit-correction-based immersed boundary–lattice Boltzmann method with two relaxation times," *Physical Review E*, vol. 89, 2014. doi:10.1103/physreve.89.023307